

Digital Twin: A Gaussian Splatting Prototype for Student Accommodation Service Decision Support

Research Motivation and Prototype

Student accommodation is both a residential environment and a service setting, in which spatial layout, and functional cues shape prospective residents' evaluations and current residents' satisfaction. Servicescape research shows that physical surroundings influence customer responses (Bitner, 1992), while student-housing evidence identifies physical, social, and management attributes as important predictors of residential satisfaction (Amole, 2009). Yet accommodation providers still rely heavily on static photographs, floor plans, and fragmented operational metrics. These tools provide either visual representation or management data, but rarely connect a realistic model of communal space to interpretable service actions.

Student Accommodation Service Application

To address this gap, we develop a browser-based digital-twin prototype of a student-accommodation communal room. The room was captured with a smartphone and reconstructed through 3D Gaussian Splatting, which supports high-quality novel-view rendering at interactive rates (Wu et al., 2024). Five calibrated overlays, i.e., temperature and comfort, occupancy, lighting, communal appliance state, and viewer engagement, are embedded at meaningful spatial locations. Environmental values follow seeded, bounded simulation with realistic noise; engagement is an interaction-derived navigation proxy rather than biometric or emotion analysis. A transparent rule layer converts signal combinations into actions: ventilate when the room is warm and occupied, improve lighting when presentation is weak, review a non-essential water boiler when the room is vacant, and focus the social zone when navigational attention is high.

Contribution and Limitations

Digital-twin research emphasizes the integration of physical and virtual entities, state data, and twinning processes (Jones et al., 2020), while built-environment scholarship highlights sensor-linked representations for monitoring and lifecycle decision support (Boje et al., 2020). Because the present signals are simulated rather than physically sensor-linked, the system should be interpreted as a digital-twin prototype rather than a fully synchronized operational twin. Nevertheless, it demonstrates a coherent service architecture for university residences and private purpose-built student accommodation. Prospective students and parents can inspect communal space remotely; accommodation teams can assess viewing readiness and comfort; facilities managers can identify avoidable energy use; and marketing teams can determine which spatial features merit stronger emphasis. This integration extends the student-accommodation servicescape across digital and physical touchpoints. The proof of concept remains limited to one room, demonstrative thresholds, and no resident or manager evaluation. Future research should integrate real IoT sensors and test effects on accommodation choice, residential satisfaction, operational efficiency, and service communication.

References

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